

Breakdown of Bell Factorization from Non-Injective Effective Descriptions

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Motivation

- Bell inequalities are experimentally violated
- Standard interpretations:
 - Nonlocality
 - Abandonment of realism
- But Bell inequalities constrain probabilistic structure

Key point: Bell constrains factorizability, not dynamics

Key Question

- Why should observable variables be factorizable?
- Standard assumption:
 - Existence of a global underlying description
 - Simultaneous assignment of all outcomes

Question: What if this assumption fails structurally?

Observable states arise from a projection:

$$\Pi : \Omega \rightarrow \mathcal{O}$$

- Ω : underlying configuration space
- $\mathcal{O}$: observable outcomes

Assumption: Π is non-injective

Multiple configurations map to the same observable outcome

- Observable outcomes are not single configurations
- They correspond to equivalence classes

Interpretation:

- Information is lost under projection
- Observable variables are coarse-grained

Bell Factorization

Bell assumes:

$$P(A, B \mid \lambda) = P(A \mid \lambda) P(B \mid \lambda)$$

- λ : underlying state
- Requires a joint assignment of all outcomes

Implicit assumption: existence of a single consistent λ

Where It Breaks

- Observable outcomes arise from multiple configurations
- Different measurements may correspond to different λ

Result:

- No global assignment of all observables
- Factorization cannot be consistently defined

Key Result

- Non-injective projection prevents simultaneous assignment
- Bell factorizability assumption does not apply

Conclusion:

- Bell inequalities may fail without nonlocal dynamics

Classical

One underlying state $\rightarrow$ one observable description

Global assignment exists $\rightarrow$ factorization

Non-injective

Multiple underlying states $\rightarrow$ one observable

No global assignment $\rightarrow$ no factorization

- Explicit construction
- Produces Bell–CHSH violations
- Preserves:
 - No-signalling
 - Measurement independence

Violations arise from projection structure

What This Does Not Claim

- No modification of quantum mechanics
- No additional hidden-variable dynamics
- No superluminal influence

What This Shows

- Factorization is not structurally guaranteed
- Non-injectivity provides a sufficient mechanism

Bell violations can arise from description alone

- When projection becomes approximately injective:
 - Underlying configurations become distinguishable
 - Global assignment is approximately restored

Classical factorization emerges as a limit

Conceptual Message

- Bell inequalities constrain factorizability
- Not necessarily locality or realism

Non-factorizability can be structural

Conclusion

- Non-injectivity $\Rightarrow$ no global assignment
- No assignment $\Rightarrow$ no factorization
- No factorization $\Rightarrow$ Bell violations

Purely structural origin

Discussion

- Relation to contextuality
- Relation to quantum probability
- Operator-algebraic formulation
- Extensions within Cosmochrony